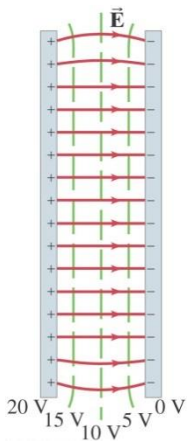


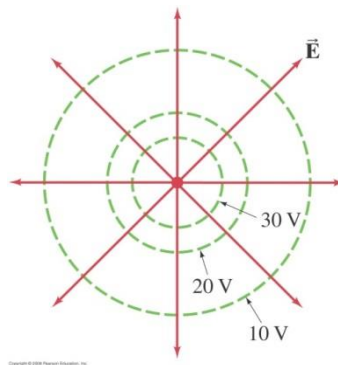


Today is cruel and tomorrow is crueller. The day after tomorrow will be wonderful; however, most people die on tomorrow night and won't see the sunshine the day after tomorrow.

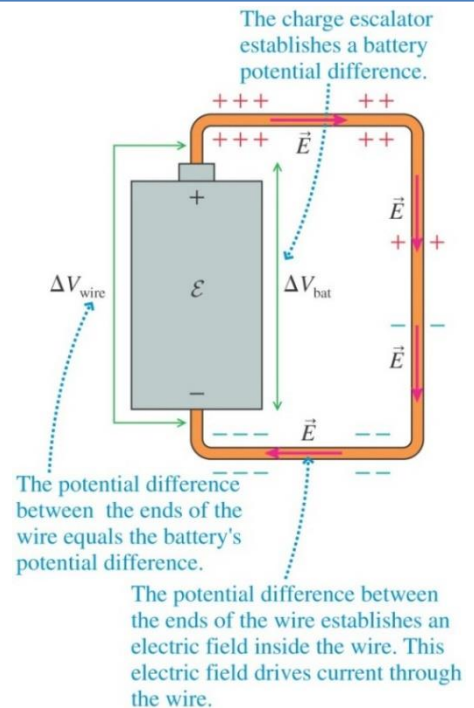
— From Ma Yun, CEO of Alibaba



Parallel Plates



Point Charges



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General $\Delta U(x) = U_2 - U_1 = - \int_1^2 F dx$

Point Charges

$\vec{F}_e = Q_2 \vec{E}$	$F = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2}$	$U = \frac{Q_1 Q_2}{4\pi\epsilon_0} \frac{1}{r}$	$U = Q_2 V$
	$E = \frac{Q_1}{4\pi\epsilon_0 r^2}$	$V = \frac{Q_1}{4\pi\epsilon_0} \frac{1}{r}$	

General

General

$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \dots$

$E_r = - \frac{\partial V}{\partial r}$

$V = V_1 + V_2 + \dots$

Self-check – after completing the lessons (lecture and tutorial) for this topic, are you able to

1. Calculate electric field in problems and demonstrate understanding of the concept of electric force, field and flux; the relationship electric potential energy and electric force
2. calculate electric potential, electric potential energy and solve problems using energy conservation

Electric force and electric potential energy

Earlier we learnt that the change in potential energy associated with a particular conservative force $\vec{F}$ as the negative of the work done by that force:

$$\Delta U = U_2 - U_1 = - \int_1^2 \vec{F} \cdot d\vec{l} = -W$$

Remembering that integration and differentiation are inverse operations"

$$\frac{d}{dx} \int F(x) dx = F(x)$$

Thus,

$$F(x) = - \frac{dU(x)}{dx} \quad \text{General}$$

The same is applicable to the charges and electric field. Previously, it was mass and the gravitational field.

Thus, electric potential energy can be written as

$$\Delta U_E = U_2 - U_1 = - \int_1^2 \vec{F}_E \cdot d\vec{l} \quad \text{General}$$

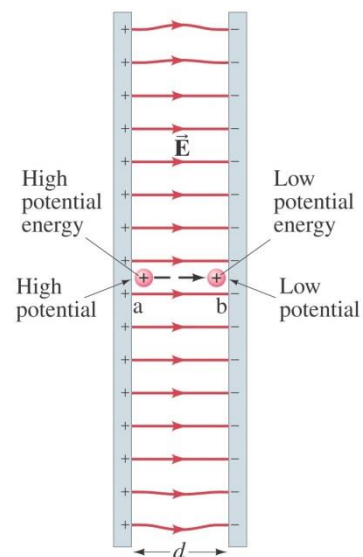
We will consider the simplest electric field, the uniform field between two parallel plates where the electric field is a constant value E anywhere in between the plates.

$$\Delta U_E = U_b - U_a = - \int_a^b \vec{F}_E \cdot d\vec{l}$$

$$0 - U_a = - \int_a^b qE dx \cos 0^\circ$$

Parallel Plates

As in gravitational PE, we need to define a point as zero PE. If the charge is -ve instead, the change in electrical potential energy moving from a to b will be



Parallel Plates

Electric potential

Electric potential is defined as electrical potential energy per unit charge:

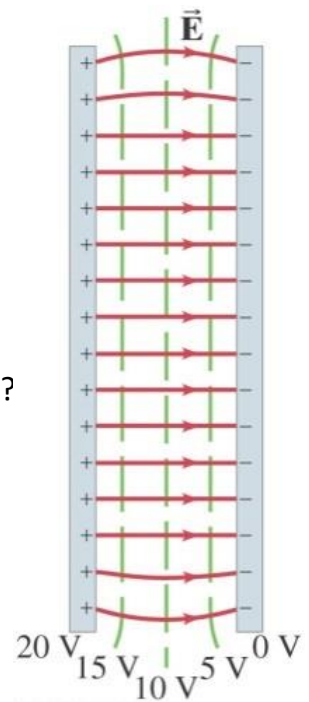
$$V = \frac{U_E}{q} \quad \text{General}$$

Unit of electric potential: the volt (V):

$$1 \text{ V} = 1 \text{ J/C.} \quad \text{General}$$

A simple numerical example:

If the separation of the two parallel plate is $d = 10 \text{ cm}$,
what is the electric potential at a point 3 cm from the positive plate?



Parallel Plates

What is the electric potential energy of an object with charge $q = 2 \text{ nC}$ at this point?

If the mass of the object is 0.01 kg, what is its velocity just before it hits the -ve plate?

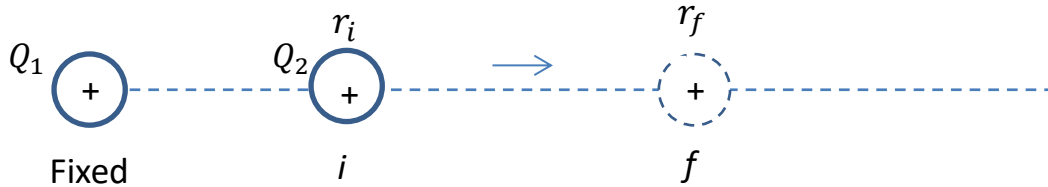
Electric potential energy 2

Consider two charges $+Q_1$ and $+Q_2$ at a distance r_i apart. The position of one of the charge is fixed while the other is free to move starting from rest at point i . A repulsive force of magnitude

$$\vec{F} = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} \hat{r}$$

Point Charges

acts between the protons while the free proton moves to a new position r_f .



The change in potential energy of charge Q_2 in the electric field of the fixed charge Q_1 is given by

$$\Delta U_E = - \int_{r_i}^{r_f} \vec{F} \cdot d\vec{r} = - \int_{r_i}^{r_f} |\vec{F}| |d\vec{r}| \cos 0^\circ$$

General

$$= - \int_{r_i}^{r_f} \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} dr$$

$$= - \frac{Q_1 Q_2}{4\pi\epsilon_0} \int_{r_i}^{r_f} \frac{1}{r^2} dr = - \frac{Q_1 Q_2}{4\pi\epsilon_0} \left[-\frac{1}{r_f} - \left(-\frac{1}{r_i} \right) \right]$$

$$= \frac{Q_1 Q_2}{4\pi\epsilon_0} \left[\frac{1}{r_f} - \frac{1}{r_i} \right] =$$

Point Charges

As we would expect because the loss in electrical potential energy is transformed to kinetic energy. If we fix the electrical potential energy at $r_f = \infty$ as $U_E = 0$, then

$$U_f - U_i = \frac{Q_1 Q_2}{4\pi\epsilon_0} \left[\frac{1}{r_f} - \frac{1}{r_i} \right]$$

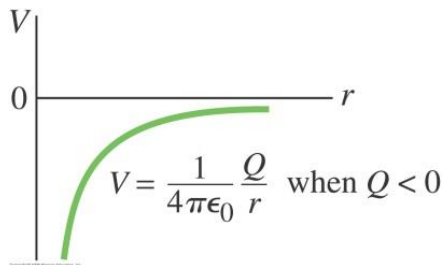
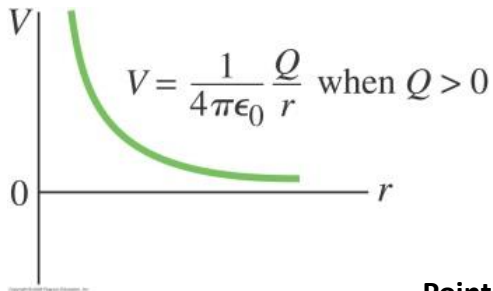
Point Charges

$\Rightarrow$ for a general r for **Point Charges**, $U = \frac{Q_1 Q_2}{4\pi\epsilon_0 r}$. As before, we can define the electric potential at a distance r from of a point charge Q as

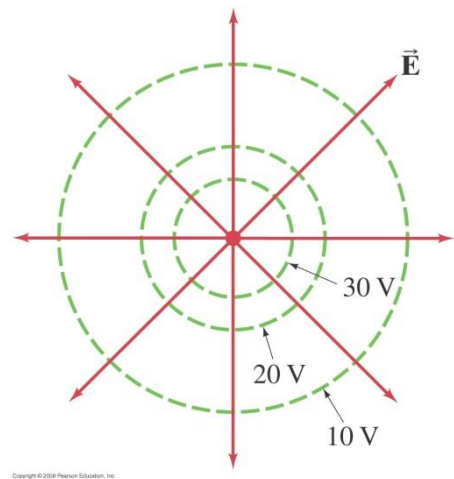
$$V = \frac{U}{q} = \frac{Q}{4\pi\epsilon_0 r}$$

Point Charges

Electric Field of Point charges



Point Charges



Giancoli pg 616, fig 23.17
The diagram shows the electric field lines and the equipotential lines of a positive point charge

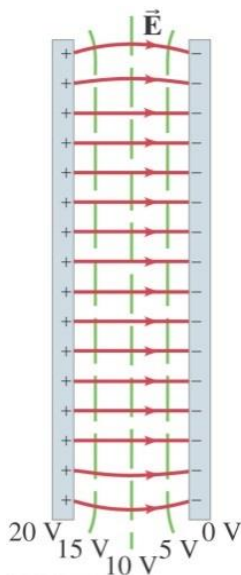
Giancoli pg 613, fig 23.10/11
The electric potential of a charge Q as a function of distance r .

Forces, Fields are vectors / scalars

General

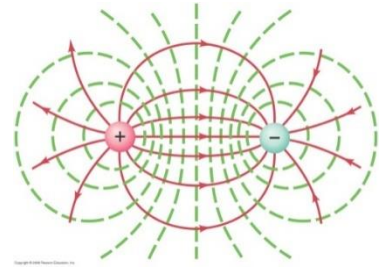
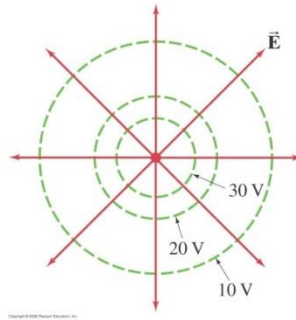
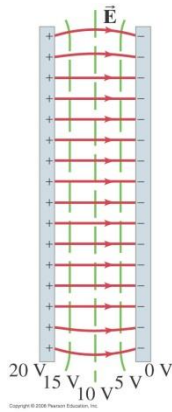
Potential energy, potential are scalars

Uniform Electric Field



Parallel Plates

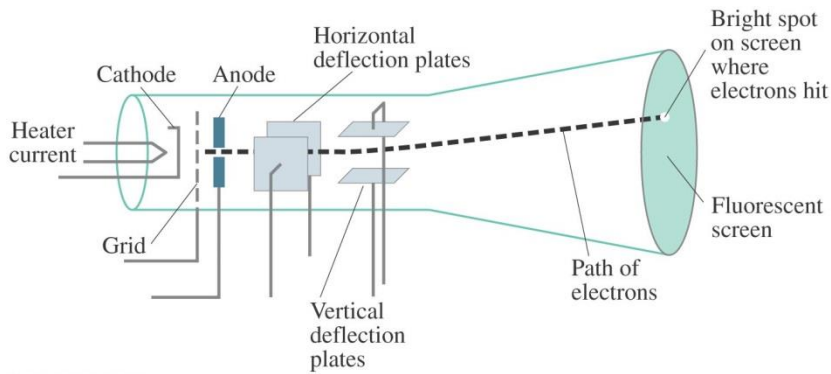
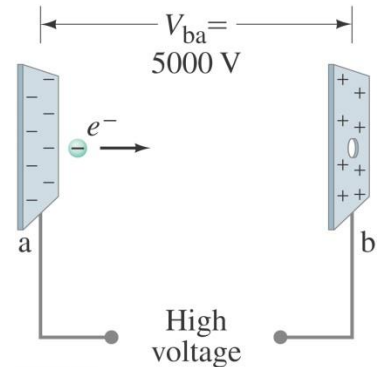
Electric Field and Equipotential Lines



	Parallel-plate	Point charge	Dipole
Electric Field			
Electric Potential			

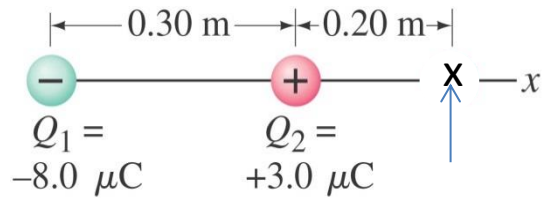
Giancoli page 610, Example 23-2: Electron in CRT.

Suppose an electron in a cathode ray tube is accelerated from rest through a potential difference $V_b - V_a = V_{ba} = +5000 \text{ V}$. (a) What is the change in electric potential energy of the electron? (b) What is the speed of the electron ($m = 9.1 \times 10^{-31} \text{ kg}$) as a result of this acceleration?

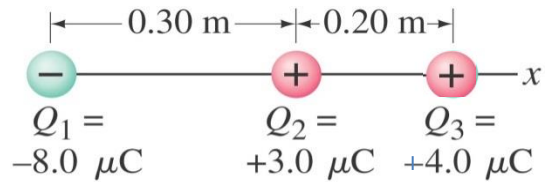


Modified Giancoli pg 566 Example 21-2: Two charges in a line

Two charged particles are arranged in a line, as shown. Calculate the electric potential at the point mark "x".



If a charge Q_3 ($= +4.0 \mu\text{C}$) is placed at "x", calculate the electric potential energy of Q_3 .

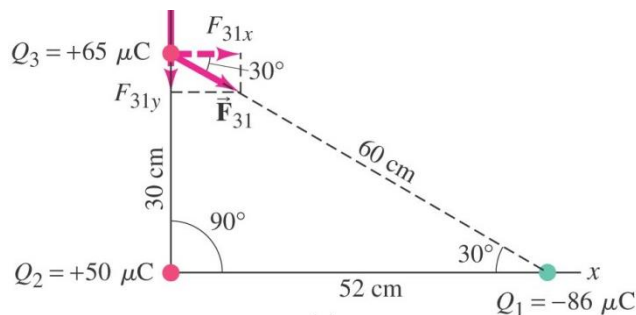


The particle with Q_3 , mass $3.2 \times 10^{-14} \text{ kg}$, is released from rest. What is its velocity when it is 0.50 m from Q_2 ?

$$2.6 \times 10^6 \text{ m/s}$$

Modified Giancoli pg 566 Example 21-3:

Calculate the electric potential energy of charge Q_3 shown in the figure due to the charges Q_1 and Q_2 .

**Electric potential V**

We can also write the vectorial form of the definition for two points in 3-D space

$$V_b - V_a = - \int_a^b \vec{E} \cdot d\vec{s} = - \int_a^b (E_x dx + E_y dy + E_z dz)$$

Conversely, given the potential $V(x, y, z)$, we can obtain the components of the electric field

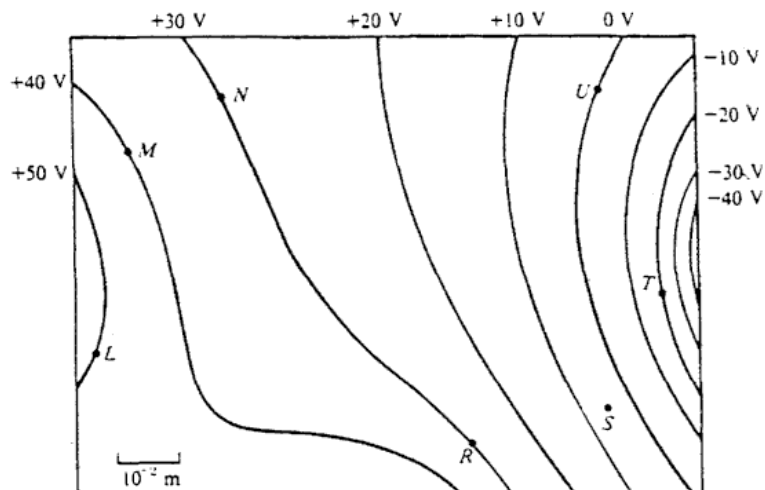
$$E_x = -\frac{\partial V}{\partial x}; E_y = -\frac{\partial V}{\partial y}; E_z = -\frac{\partial V}{\partial z}$$

Here, $\frac{\partial V}{\partial x}$ denotes a partial derivative – differentiating w.r.t. x and treating variables y and z in V as constants. It applies similarly to $\frac{\partial V}{\partial y}$ and $\frac{\partial V}{\partial z}$. If the electric field only varies in the radial direction with r , then

$$E_r = -\frac{\partial V}{\partial r}.$$

The negative sign is associated with the fact that the electric field is always pointing from higher potential to lower potential. We note that another unit for the electric field is volts per metre (V/m).

Electric Field and Electric Potential

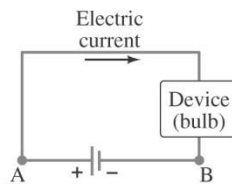
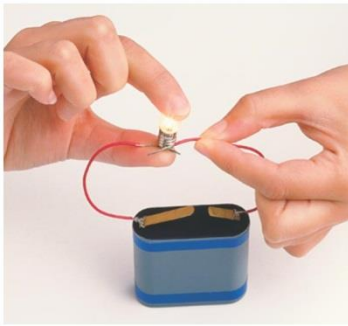


http://dev.physicslab.org/Document.aspx?doctype=3&filename=Electrostatics_PointChargesEPE.xml

- i. Where is the electric field strongest? L, M, N, R, S, T, U
- ii. Where is the electric field weakest? L, M, N, R, S, T, U
- iii. What is the direction of the electric field at R?
- iv. How much work would it take an external agent to move a charge $2 \mu\text{C}$ from R to N?
- v. Does it take more work to move a $2 \mu\text{C}$ charge from R to L and then to T compared to going directly to T?
- vi. How much EPE did the $2 \mu\text{C}$ charge have while it was at rest at position R?
- vii. What is the $2 \mu\text{C}$ charge's EPE at point T?
- viii. How much work was required to move the $2 \mu\text{C}$ charge from R to T?

"I have not failed. I've just found 10,000 ways that won't work."

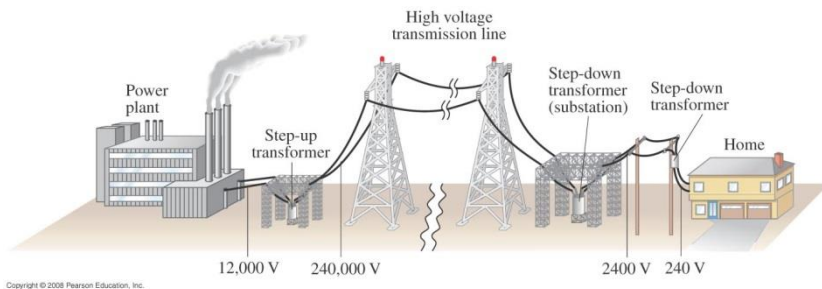
- Thomas Alva Edison (1847-1931)



$$I = \frac{dQ}{dt}; \quad I = n_e e v_d A$$

$$V = IR$$

$$R = \frac{\rho l}{A}; \quad \sigma = \frac{1}{\rho}$$



$$U = Q\Delta V$$

$$P = I\Delta V$$

$$P = \frac{(\Delta V)^2}{R} = I^2 R$$

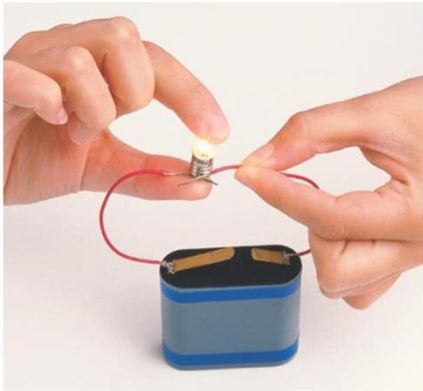
Self-check – after completing the lessons (lecture and tutorial) for this topic, are you able to

1. demonstrate understanding of flow of electrical current and how quantities such as resistance (and the properties affecting it) and electric potential difference affects current.
2. analyze the electric power of systems and implications on design of such systems
3. analyze systems with alternating currents, such as determining root mean square values of quantities
4. solve electrical circuits with resistors and/or capacitors using Kirchhoff laws
5. design and analyze basic circuits with resistors and/or capacitors for various applications

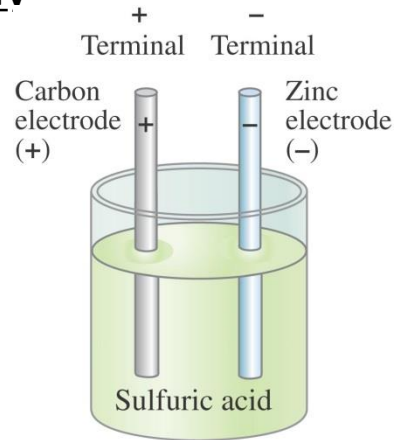
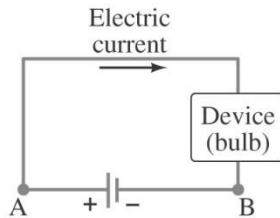
Electrical Cells / Battery

Volta discovered that electricity could be created if dissimilar metals were connected by a conductive solution called an electrolyte. This is a simple electric cell.

A battery transforms chemical energy into electrical energy.



Giancoli page 654, Fig 25.6



Giancoli page 653, Fig 25.3

Electric current I

Electric current I is defined as the rate at which electric charge flows past a given cross section A . If ΔQ is the amount of charges that flows past this surface of area A over a time Δt , then the average current is defined as

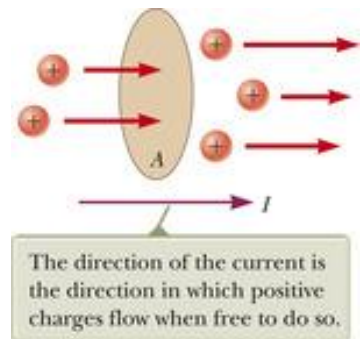
$$I_{avg} = \frac{\Delta Q}{\Delta t}.$$

If the current is time varying, the instantaneous current I is taken in the limit $\Delta t \rightarrow 0$,

$$I = \frac{dQ}{dt}.$$

The S.I. unit of current is ampere (A):

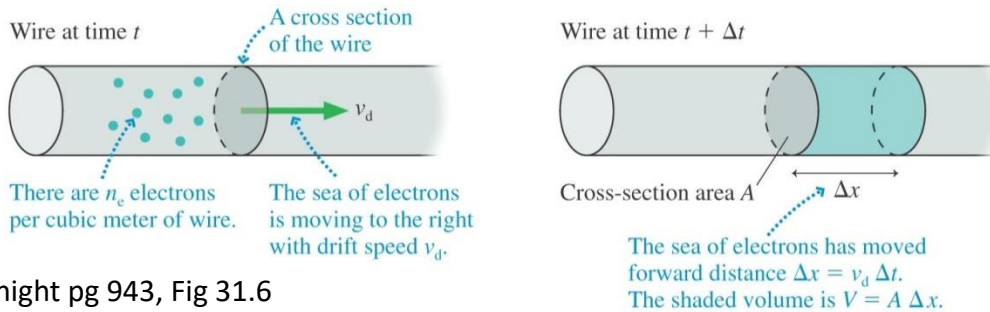
$$1\text{A} = 1\text{C/s}$$



Serway pg 772, Fig 27.1

Microscopic model of electrical conduction

The conduction electrons in a metal, like gas molecules in a vessel, are in random motion. Collectively, there is no net motion. However, if an electric field is applied, the sea of electrons will drift against the direction of electric field. This velocity of this net motion is known as the drift velocity, v_d .



Knight pg 943, Fig 31.6

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For a wire made of a material with number density of conduction electron n_e and cross sectional area A , if the drift velocity is v_d , the total charge Q passing a cross section on the wire in a time interval Δt is

$$Q = n_e (v_d \Delta t) A e.$$

Hence, the electric current, charge flow per unit time, is given by

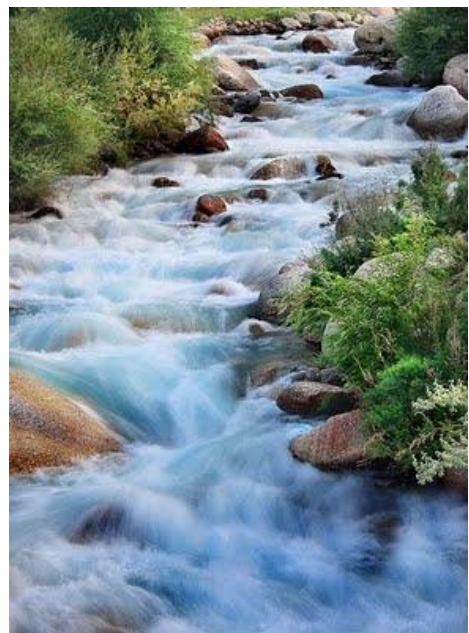
$$I = n_e e v_d A.$$

Charge carriers can be positive or negative but conventional current is defined in the direction of the flow of positive charges.

Another useful quantity to define is the current flow per unit area, the current density

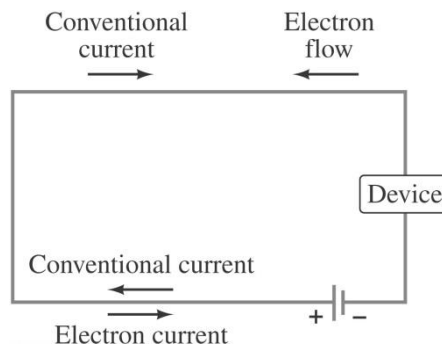
$$J = \frac{I}{A} = n_e e v_d$$

The S.I. unit for current density is A/m^2 .



Giancoli pg 654, Example 25-1: Current is flow of charge.

A steady current of 2.5 A exists in a wire for 4.0 min. (a) How much total charge passed by a given point in the circuit during those 4.0 min? (b) How many electrons would this be?



Example [Serway pg 774, Ex 27.1]

The 12-gauge copper wire in a typical residential building has a cross sectional area of $3.31 \times 10^{-6} \text{ m}^2$. It carries a constant steady current of 10.0 A. What is the drift speed of the electrons in the wire? Assume each copper atom contributes one free electron. The density of copper is 8.92 g/cm^3 . Checking the periodic table, we find that 1 mole of copper atoms, i.e. 6.02×10^{23} copper atoms, have a total mass of 63.5 g.

To estimate electron number density n_e , we can estimate first the number of copper atoms in 1 m^3 of wire.

Number of copper atoms in 1 m^3 of copper:

$$\left(\frac{8.92}{63.5} \times 6.02 \times 10^{23} \right) \times (10^2)^3 =$$

Each copper atom contributes 1 electron.

Ohm's Law: Resistance and Resistors

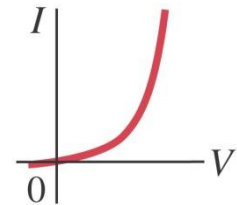
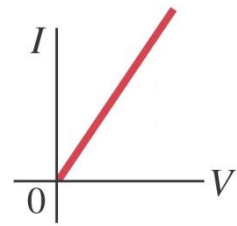
Experimentally, it is found that the current in a wire is proportional to the potential difference between its ends:

$$I \propto V$$

The ratio of voltage to current is called the resistance:

$$I = \frac{V}{R}$$

$$V = IR$$



In many conductors, the resistance is independent of the voltage; this relationship is called Ohm's law. Materials that do not follow Ohm's law are called nonohmic.

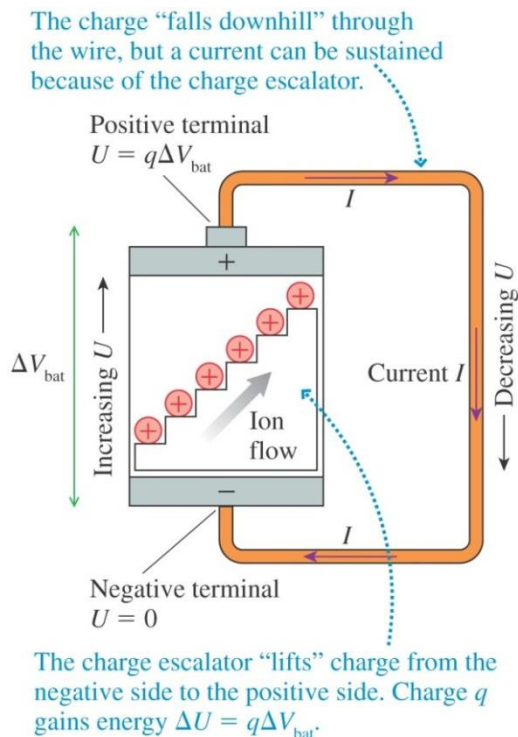
Google homework: Superconductors

Giancoli pg 657 Example 25-4: Flashlight bulb resistance.

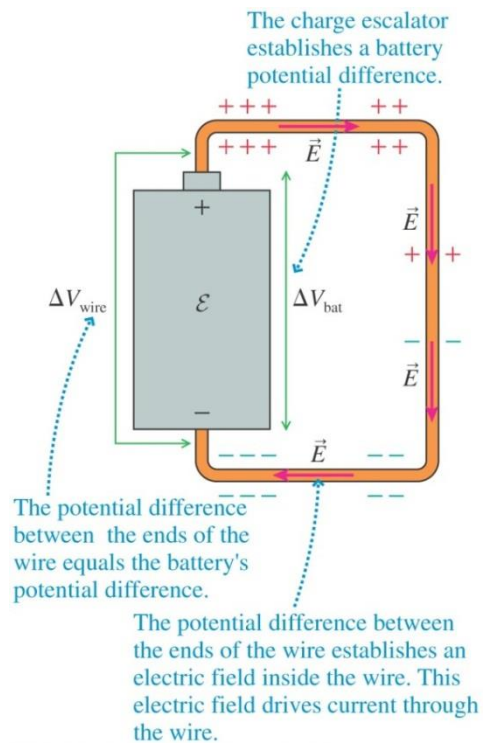
A small flashlight bulb draws 300 mA from its 1.5-V battery. (a) What is the resistance of the bulb? (b) If the battery becomes weak and the voltage drops to 1.2 V, how would the current change?

An E.M.F. source

We will now learn about ideal E.M.F. sources (batteries with no internal resistances) and model the movement of the charge carriers in the circuit.



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[Taken from Knight pg 958; Fig 31.20]

The key ideas here are

1. The battery is a source of potential difference between the ends of the wire.
2. The current is in the direction of decreasing potential (+ -> -).
3. Batteries DO NOT produce a constant current, rather batteries maintain a constant potential difference, or nearly so.
4. Electric current passes through a wire or device, and the magnitude depends on that device's resistance.

Types of Batteries:



<https://www.youtube.com/watch?v=DzTlyM-IA-I>

Important Batteries Concepts

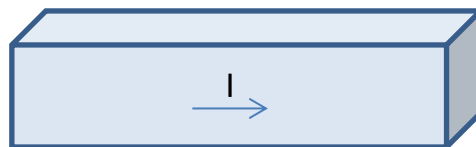


<https://www.youtube.com/watch?v=cxkVxi9P0EA>

Resistance and Resistivity

The resistance of a wire is directly proportional to its length and inversely proportional to its cross-sectional area:

$$R = \frac{\rho l}{A}$$



The constant ρ , the resistivity, is characteristic of the material, A the cross sectional area and l is the length of the wire.

The reciprocal of resistivity is conductivity $\sigma = \frac{1}{\rho}$.

TABLE 25–1 Resistivity and Temperature Coefficients (at 20°C)

Material	Resistivity, ρ ($\Omega \cdot \text{m}$)	Temperature Coefficient, α ($^{\circ}\text{C}^{-1}$)
<i>Conductors</i>		
Silver	1.59×10^{-8}	0.0061
Copper	1.68×10^{-8}	0.0068
Gold	2.44×10^{-8}	0.0034
Aluminum	2.65×10^{-8}	0.00429
Tungsten	5.60×10^{-8}	0.0045
Iron	9.71×10^{-8}	0.00651
Platinum	10.60×10^{-8}	0.003927
Mercury	98.00×10^{-8}	0.0009
Nichrome (Ni, Fe, Cr alloy)	100.00×10^{-8}	0.0004
<i>Semiconductors</i> [†]		
Carbon (graphite)	$(3 - 60) \times 10^{-5}$	−0.0005
Germanium	$(1 - 500) \times 10^{-3}$	−0.05
Silicon	0.1 – 60	−0.07
<i>Insulators</i>		
Glass	$10^9 - 10^{12}$	
Hard rubber	$10^{13} - 10^{15}$	

[†] Values depend strongly on the presence of even slight amounts of impurities.

Giancoli pg 659 Example 25-5: Speaker wires.

Suppose you want to connect your stereo to remote speakers. (a) If each wire must be 20 m long, what diameter copper wire should you use to keep the resistance less than $0.10\ \Omega$ per wire? (b) If the current to each speaker is 4.0 A, what is the potential difference, or voltage drop, across each wire?

$$\rho_{cu} = 1.68 \times 10^{-8} \Omega m$$

Giancoli pg 659 Example 25-6: Stretching changes resistance.

Suppose a wire of resistance R could be stretched uniformly until it was twice its original length. What would happen to its resistance?

Temperature Dependence of Resistivity

For any given material, the resistivity increases with temperature:

$$\rho_T = \rho_o(1 + \alpha(T - T_o))$$

Semiconductors are complex materials, and may have resistivities that decrease with temperature. This is because more charge carriers become mobile when the temperature is higher and are no longer bounded to local lattice ions.

Giancoli pg 660, Example 25-7: Resistance thermometer.

The variation in electrical resistance with temperature can be used to make precise temperature measurements. Platinum is commonly used since it is relatively free from corrosive effects and has a high melting point. Suppose at 20.0°C the resistance of a platinum resistance thermometer is 164.2 Ω . When placed in a particular solution, the resistance is 187.4 Ω . What is the temperature of this solution?

$$\rho_T = \rho_o(1 + \alpha(T - T_o))$$

$\times \frac{l}{A}$:

$$\frac{\rho_T l}{A} = \frac{\rho_o l}{A} (1 + \alpha(T - T_o))$$
$$R_T = R_o(1 + \alpha(T - T_o))$$

$$T =$$

Electrical power

The rate at which electrical energy of a system decreases as charge Q passes through a resistor

$$\frac{dU}{dt} = \frac{d}{dt}(Q\Delta V) = \frac{dQ}{dt}\Delta V = I\Delta V.$$

Hence, the power delivered to the resistor is

$$P = I\Delta V$$

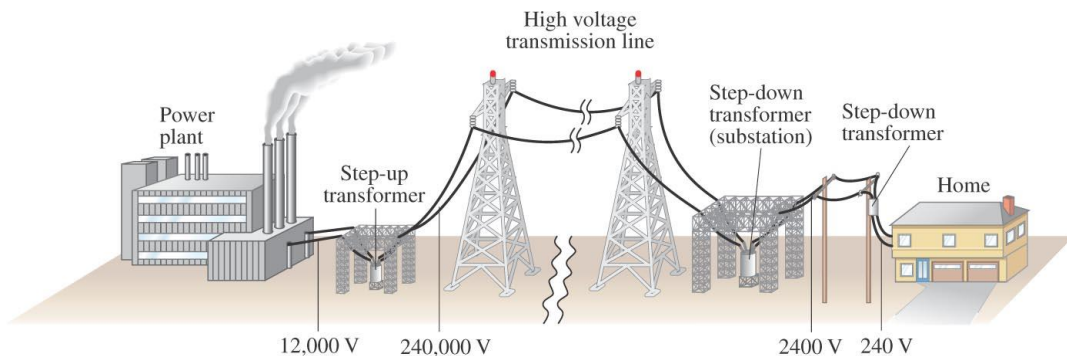
or since $\Delta V = IR$,

$$P = \frac{(\Delta V)^2}{R} = I^2 R.$$



<https://www.youtube.com/watch?v=nbPmsBmo03Y&t=4s>

Electrical power transmission



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Giancoli pg 772 Fig 29.26

Giancoli pg 772 Example 29-13: Transmission lines.

An average of 120 kW of electric power is sent to a small town from a power plant 10 km away. The transmission lines have a total resistance of $0.40 \, \Omega$. Calculate the power loss if the power is transmitted at (a) 240 V and (b) 24,000 V.

Alternating Current

Current from a battery flows steadily in one direction (direct current, DC).
Current from a power plant varies sinusoidally (alternating current, AC).

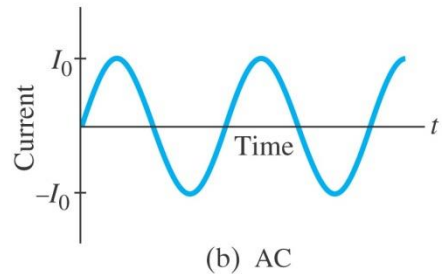
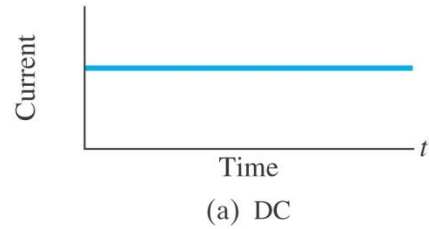
The voltage varies sinusoidally with time:

$$V = V_o \sin 2\pi ft = V_o \sin \omega t$$

So does the current

$$I = \frac{V}{R} = \frac{V_o}{R} \sin 2\pi ft = I_o \sin \omega t$$

So how does electrical power $P = I^2 R$ vary with time?



$$P = I^2 R = I_o^2 \sin^2 \omega t$$

Consider an electric kettle that runs on A.C. with voltage and current given by the graphs above.



The total average power is given by

$$I_{rms} = \frac{I_o}{\sqrt{2}}; V_{rms} = \frac{V_o}{\sqrt{2}}; P_{avg} = I_{rms} V_{rms}$$

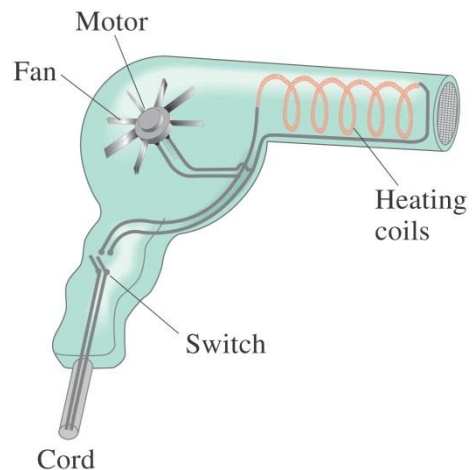
If the A.C. is to be replaced by D.C. such that the water takes the same time to boil, what should the D.C. be?

The root-mean-square current I_{rms} and voltage V_{rms} of a sinusoidal A.C. are given by

The average power of a sinusoidal A.C. is given by

Giancoli pg 655 Example 25-13: Hair dryer.

(a) Calculate the resistance and the peak current in a 1000-W hair dryer connected to a 120-V line. (b) What happens if it is connected to a 240-V line in Britain?



Electrical Safety (Taken from Knight Chap 26.4)

The relative safety of electricity is not governed by the voltage but by the current.

Current—the flow of charges through the body—is what produces physiological effects and damage because it mimics nerve impulses and causes muscles to involuntarily contract.

TABLE 26.1 Physiological effects of currents passing through the body

Physiological effect	AC current (rms) (mA)	DC current (mA)
Threshold of sensation	1	3
Paralysis of respiratory muscles	15	60
Heart fibrillation, likely fatal	> 100	> 500



- A girl touching a Van de Graaff generator is not in danger of shock because she is standing on an insulating surface.
- The high resistance of the platform means little current is passing through her to the ground.
- Even if she touches a grounded object, the current is modest because the charge of the generator is small.



<https://www.youtube.com/watch?v=ubZuSZYVBng&t=13s>



<https://www.youtube.com/watch?v=sBUvYh93pQc&t=4s>

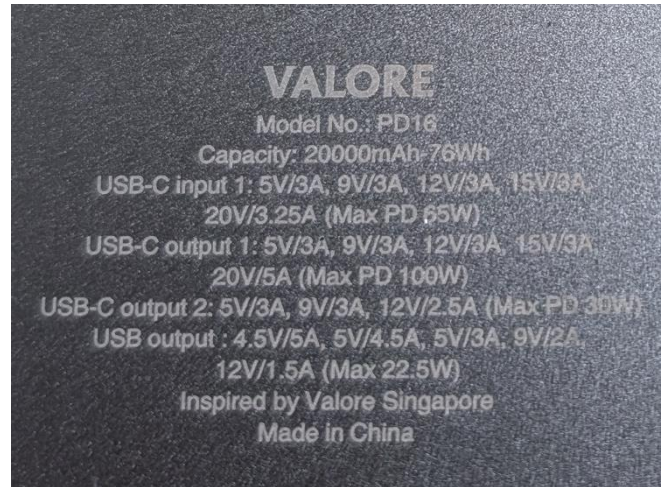


<https://www.youtube.com/watch?v=L8mIZIMNjpQ&t=3s>



<https://www.youtube.com/watch?v=4HOS-Th0ndo&t=4s>

Power Banks



What do the numbers on the power bank mean?

Appendix 1: More Microscopic model of electrical conduction

[Non-examinable]

We will next derive an expression for the drift velocity $\vec{v}_d$ and relate it to an electric field $\vec{E}$ provided by the e.m.f. source. When a free electron of mass m and with charge $q(=-e)$ is subject to an electric field $\vec{E}$, it experiences a force

$$\vec{F} = q\vec{E} = me\vec{a}$$

and an acceleration

$$\vec{a} = \frac{q\vec{E}}{m_e}.$$

For electrons moving in a wire, the average final velocity $\vec{v}_f$, after the electric field $\vec{E}$ is applied for a time t , is given by

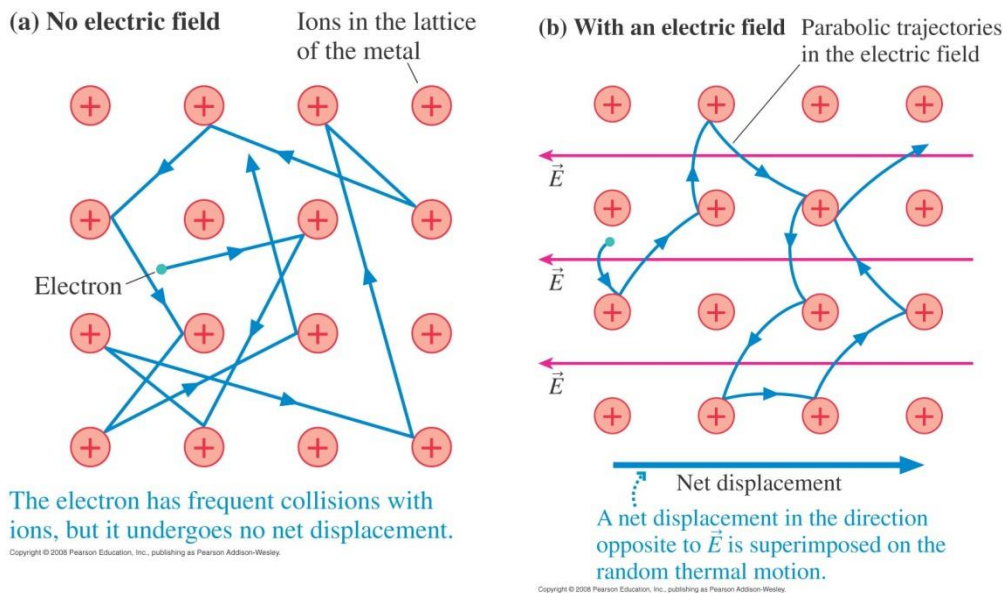
$$\vec{v}_f = \vec{v}_i + \vec{a}\tau$$

where τ is the average time between successive collisions between electrons and the positive ions. We can regard $\vec{v}_i = 0$, since the initial velocities are randomly distributed over all directions. Thus, we can write

$$\vec{v}_f = \vec{v}_d = \frac{q\vec{E}}{m_e}\tau$$

Therefore, from $I = nev_dA$

$$J = \frac{I}{A} = n_e \frac{e^2\vec{E}}{m_e}\tau.$$



Taken from Knight pg 948, Fig 31.13

Appendix 1: More Microscopic model of electrical conduction

[Non-examinable]

Continuing from our previous results

$$J = \frac{I}{A} = \frac{n_e e^2 \tau}{m_e} \vec{E}.$$

We have a simple relation that relates the current density J to electric field $\vec{E}$. This is true for some materials. These materials that follow the relation

$$J \propto \vec{E}$$

are said to follow Ohm's law. The constant of proportionality which is dependent on the properties of the material, n_e and τ is known as conductivity

$$\sigma = \frac{n_e e^2 \tau}{m_e}.$$

The S.I. unit for σ is $(\Omega \cdot m)^{-1}$. The inverse of conductivity is resistivity

$$\rho = \frac{1}{\sigma}.$$

The S.I. unit for ρ is $(\Omega \cdot m)$. Consider a segment of wire with uniform cross sectional area A and length l . We can write

$$\Delta V = El$$

Thus,

$$J = \frac{I}{A} = \sigma \frac{\Delta V}{l}$$

and

$$\Delta V = \left(\frac{l}{\sigma A} \right) I = RI$$

where we have defined resistance

$$R = \frac{\Delta V}{I} (= \frac{l}{\sigma A} = \frac{\rho l}{A}). \quad [\text{Serway pg 775, Fig 27.5}]$$

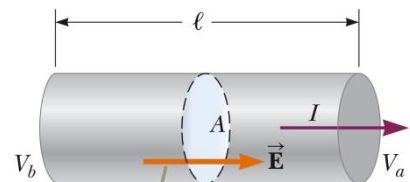
The S.I. unit for resistance is Ω (Ohm).

$$1\Omega = 1 \text{ V/A}.$$

Every ohmic material has a characteristic resistivity that depends on the properties of the material and temperature:

$$\rho = \rho_o (1 + \alpha(T - T_o))$$

where ρ_o is the resistivity taken at some reference temperature T_o and α is the temperature coefficient of resistivity.



A potential difference $\Delta V = V_b - V_a$ maintained across the conductor sets up an electric field $\vec{E}$, and this field produces a current I that is proportional to the potential difference.



2026S1 PH1012: Physics A
Kirchhoff Laws; Capacitors and Capacitance

Dr Ho Shen Yong
Lecturer, School of Physical and Mathematical Sciences
Nanyang Technological University

Week 10
Serway Chap 27, 25

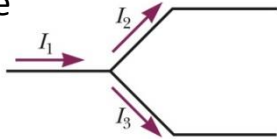
"Enthusiasm spells the difference between mediocrity and accomplishment."

- Norman Vincent Peale (1898 – 1993)

The amount of charge flowing out of the branches on the right must equal the amount flowing into the single branch on the left.

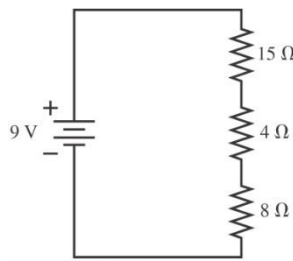
Junction Rule

$$I_1 = I_2 + I_3$$



a

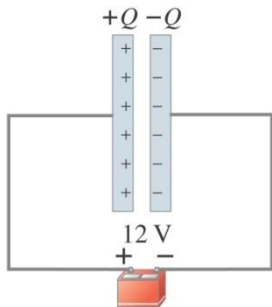
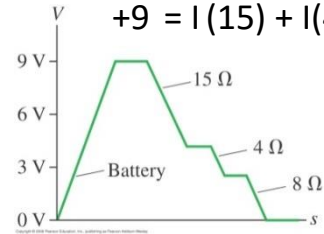
(a)



Loop Rule

$$\sum e.m.f. = \sum Ir$$

$$+9 = I(15) + I(4) + I(8)$$



Parallel

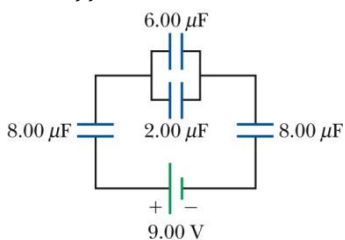
(a)

(b)

$$C_{eff} = C_1 + C_2 + \dots$$

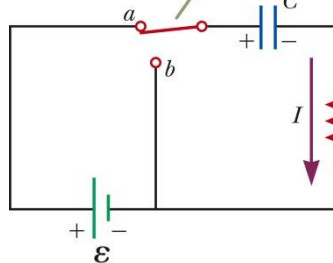
Series

$$\frac{1}{C_{eff}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$$

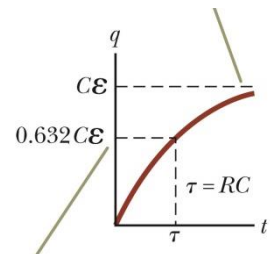


$$C = \frac{Q}{V}; C = \frac{\epsilon_0 A}{d}$$

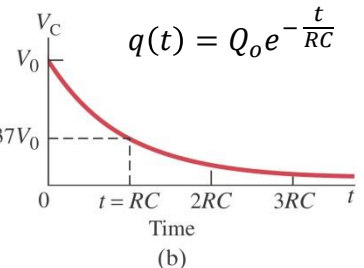
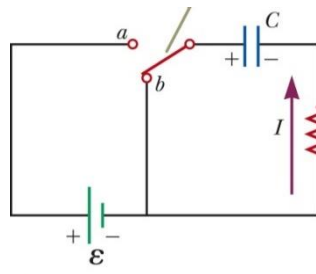
$$U = \frac{1}{2} CV^2$$



$$\tau = RC$$



$$q(t) = Q(1 - e^{-\frac{t}{RC}})$$



$$q(t) = Q_0 e^{-\frac{t}{RC}}$$

Kirchhoff's law

1. Junction Rule: At any junctions, the sum of the currents must be equal to zero.

$$\sum_{\text{junction}} I = 0$$

Conservation of charges

2. Loop Rule: The sum of the potential differences across all elements around any closed circuit loop must be zero.

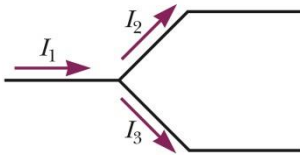
$$\sum_{\text{closed loop}} \Delta V = 0$$

Conservation of energy

Illustrations:

[Serway pg 804, Fig 28.12]

The amount of charge flowing out of the branches on the right must equal the amount flowing into the single branch on the left.

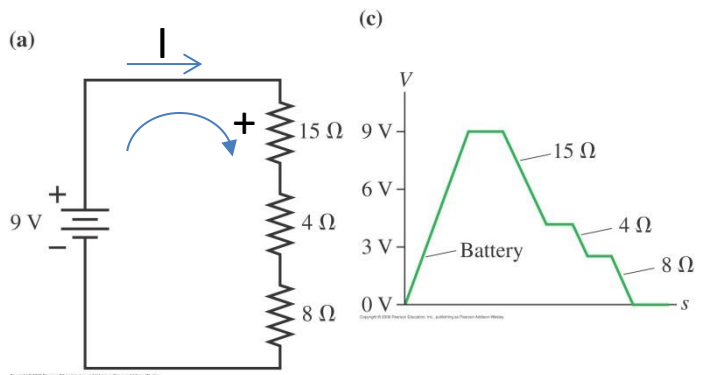


a

Junction Rule

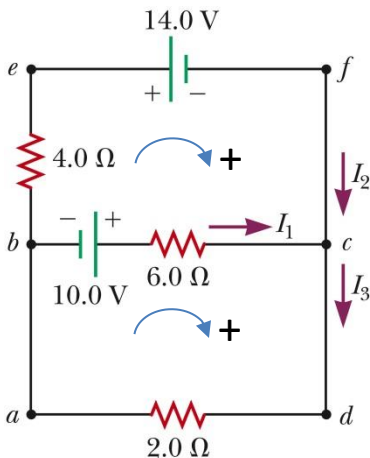
$$I_1 = I_2 + I_3$$

[Knight pg 976, Ex32.6]



Loop Rule

$$+9 = I(15) + I(4) + I(8)$$



[Serway pg 806; Ex 28.7]

$$\text{Junction c: } I_1 + I_2 = I_3$$

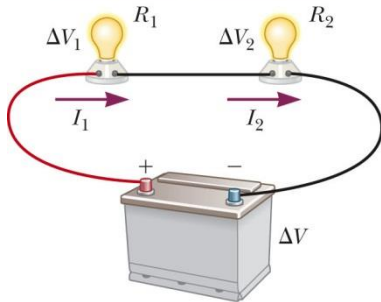
$$\text{Loop } abcd: 10 = 6I_1 + 2I_3$$

$$\text{Loop } befcb: -10 - 14 = 4I_2 - 6I_1$$

$$\text{Solve to get } I_1 = 2\text{A}; I_2 = -3\text{A}; I_3 = -1\text{A};$$

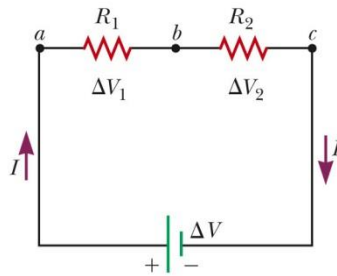
Resistors in series [Serway pg 798; Ex 28.3]

A pictorial representation of two resistors connected in series to a battery



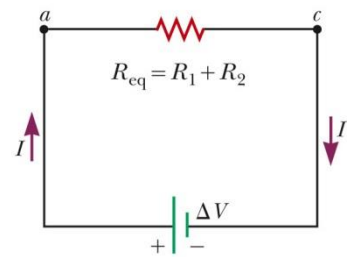
a

A circuit diagram showing the two resistors connected in series to a battery



b

A circuit diagram showing the equivalent resistance of the resistors in series



c

Resistors in series combination has the same current through them.

Using Kirchhoff's loop law:

$$\Delta V - IR_1 - IR_2 = 0$$

$$\Delta V = IR_{eq} = I(R_1 + R_2)$$

$$R_{eq} = (R_1 + R_2)$$

In general, for resistors connected in series,

$$R_{eq} = (R_1 + R_2 + \dots)$$

E.M.F. and internal resistance (Real Batteries)

The maximum possible voltage the battery can provide between its terminals is its emf $\mathcal{E}$.

However, in a real life, there is a resistance to the flow of charge within the battery – known as internal resistance r . Therefore, the terminal voltage of a battery ΔV when there is a current I is given by

$$\Delta V = \mathcal{E} - Ir.$$

The voltage drop across the load resistor is

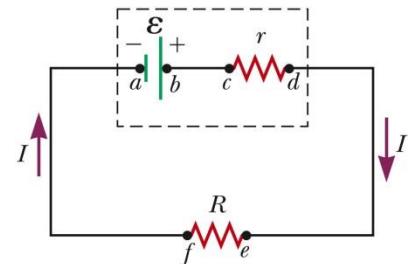
$$\Delta V = IR.$$

Thus,

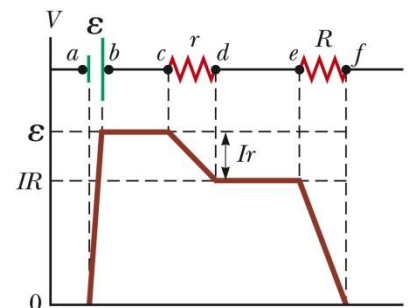
$$\mathcal{E} - Ir = IR$$

$$I = \mathcal{E}/(R+r)$$

[Serway pg 795; Ex 28.1]



a

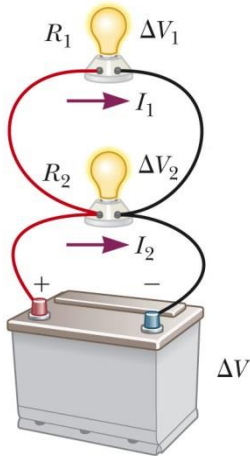


b

Resistors in parallel

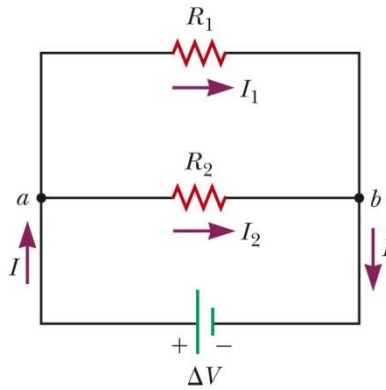
[Text pg 799; Ex 28.5]

A pictorial representation of two resistors connected in parallel to a battery



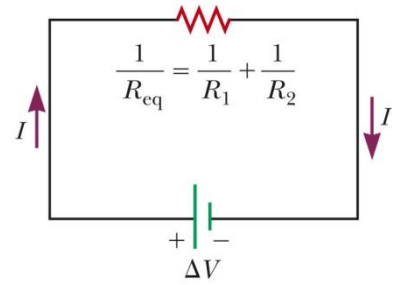
a

A circuit diagram showing the two resistors connected in parallel to a battery



b

A circuit diagram showing the equivalent resistance of the resistors in parallel



c

Resistors in parallel combination has the same potential difference across them.
Using Kirchhoff's junction law:

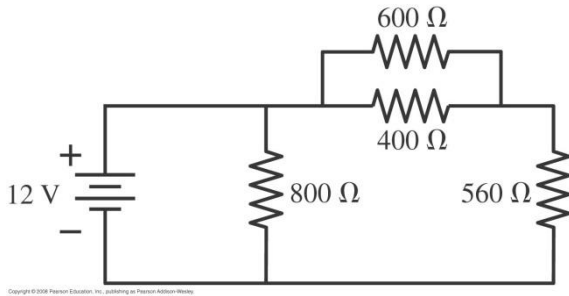
$$I - I_1 - I_2 = 0$$

$$I = \Delta V / R_{eq} = \Delta V / R_1 + \Delta V / R_2$$

In general, for resistors connected in parallel,

$$1/R_{eq} = (1/R_1 + 1/R_2 + \dots)$$

Example: [Knight pg 984; Ex 32.11]



Find the effective resistance of the resistors; and the current through each resistor and the voltage across each resistor.

Capacitance

[Text pg 741; Halliday and Resnick Chap 31; Serway]

The capacitance C of a capacitor is defined as the ratio of the magnitude of the charge (Q) on either conductor to the magnitude of the potential difference between the conductors (ΔV):

$$C \equiv \frac{Q}{\Delta V}.$$

Thus, a capacitor with large capacitance is one which can hold a huge charge even when the potential difference across the two plates is small. The S.I. unit for capacitance is farad (F).

$$1\text{F} = 1\text{C} / 1\text{V}.$$

The farad is a very large unit (because the coulomb is a very large unit; the force between two 1 C charges 1 m apart is of the order 10^{10} N!). In practice, capacitors have capacitance in the range of microfarads (μF , 10^{-6}F) to picofarads (pF , 10^{-12}F).

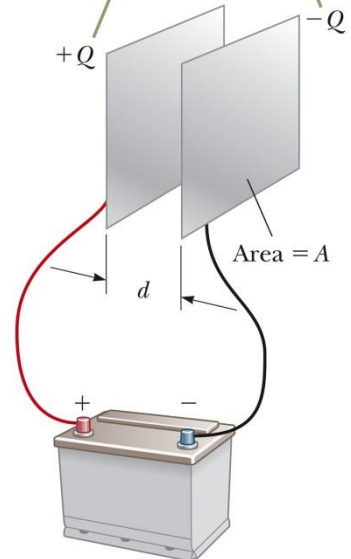
The capacitance of a capacitor is dependent on:

1. shapes and relative positions of plates;
2. the material that fills between the plates.

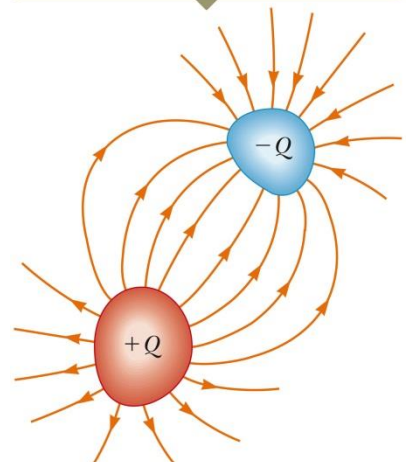
In the figure , we have a generalized capacitor:

1. Two conductors of arbitrary shape, which we will call plates regardless of their shape.
2. Charged capacitor carries equal and opposite charges $+Q$ and $-Q$. Note that the net charge of a capacitor is zero, not Q . Q represents the absolute value of charge on either plate.
3. A capacitor can be charged by connecting the two plates to opposite terminals of a battery. The plates are conductors and charges flow such that the plates are at equal potential with the terminals of the battery they are connected to.

When the capacitor is connected to the terminals of a battery, electrons transfer between the plates and the wires so that the plates become charged.



When the capacitor is charged, the conductors carry charges of equal magnitude and opposite sign.



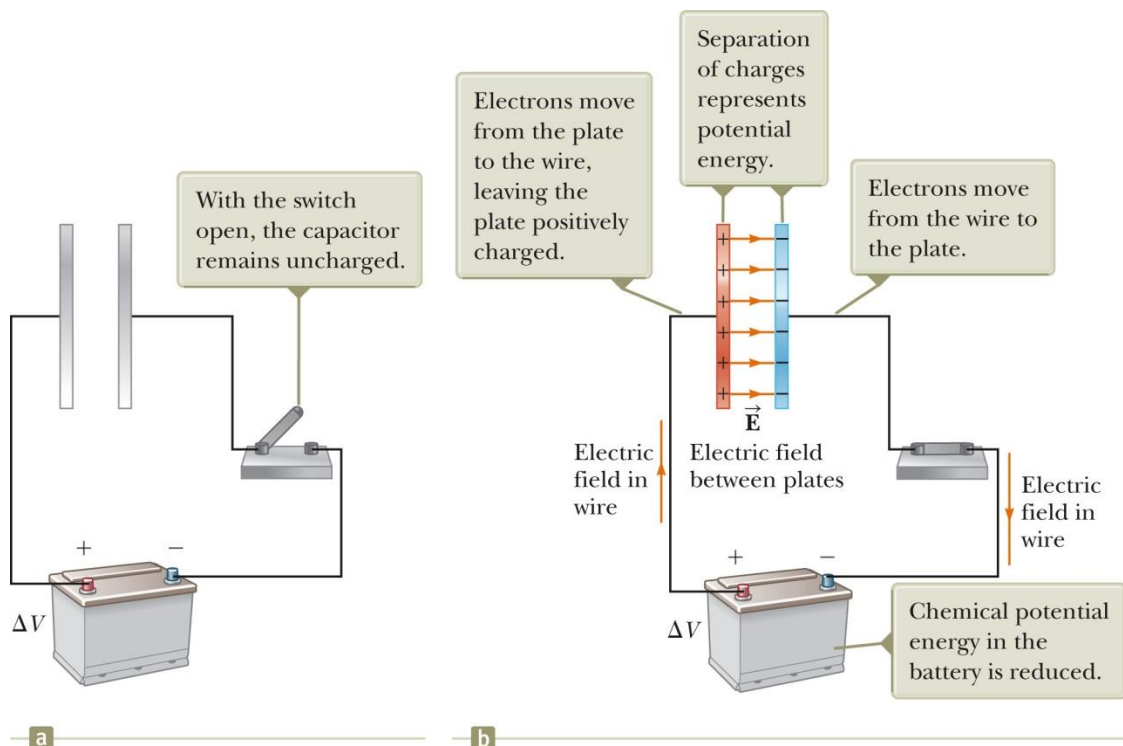
Some facts about charges on conductors

1. An excess charge placed on an isolated conductor moves entirely to the outer surface of the conductor. None of the excess charge is found within the body of the conductor.
2. An excess charge placed on an isolated conductor distributes itself on the surface so that all points of the conductor – whether on the surface or inside – come to the same potential.

A charge storing device – the capacitor

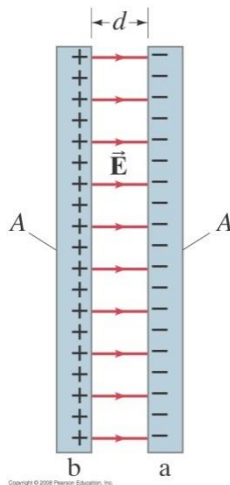
We will be studying about a charge storing device that is commonly used in circuits. Using our knowledge of electric fields and potentials, we will learn how to compute the 'charge storing ability' of a given system consisting of two conductors. The capacity of such a system to hold charges is known as capacitance.

Understanding of capacitor charging [Serway Fig 25.10]



Giancoli pg 630: Capacitance for a parallel plate capacitor

Derive the capacitance of parallel-plate capacitor with area A separated by a distance d .



$$\Delta V = - \int_a^b \vec{E} \cdot d\vec{r} = Ed = \frac{\sigma d}{\epsilon_o} = \frac{Qd}{A\epsilon_o}$$

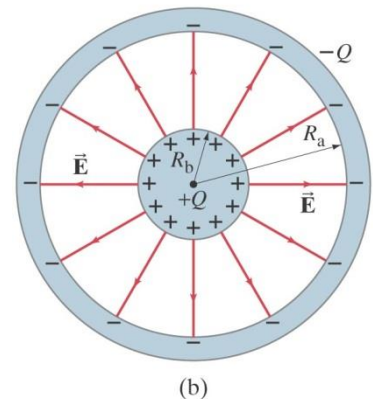
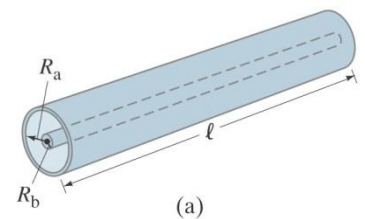
$$C = \frac{Q}{\Delta V} = \frac{\epsilon_o A}{d}$$

[Non-Examinable]

Giancoli pg 631 Example 24-2:

Capacitance for a cylindrical capacitor

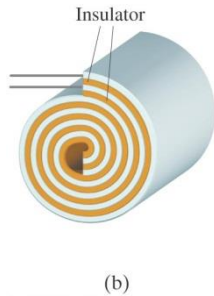
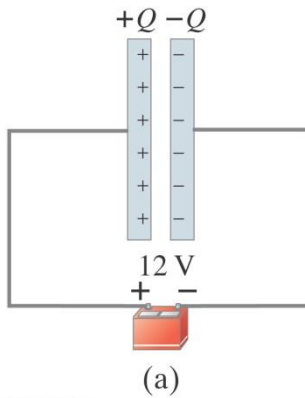
Derive the capacitance of a cylindrical capacitor that consists of a cylinder (or wire) of radius R_b surrounded by a coaxial cylindrical shell of inner radius R_a . Both cylinders have length l which we assume is much greater than the separation of the cylinders, so we can neglect end effects. The capacitor is charged (by connecting it to a battery) so that one cylinder has a charge $+Q$ (say, the inner one) and the other one a charge $-Q$. Determine a formula for the capacitance.



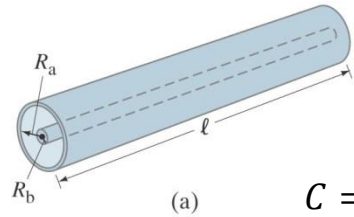
$$\Delta V = - \int_a^b \vec{E} \cdot d\vec{l} = - \int_a^b \frac{\lambda}{2\pi\epsilon_o r} dr = - \frac{Q}{2\pi\epsilon_o l} \ln\left(\frac{b}{a}\right)$$

$$C = \frac{Q}{\Delta V} = \frac{2\pi\epsilon_o l}{\ln\left(\frac{a}{b}\right)}$$

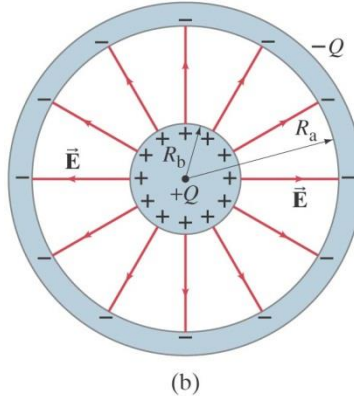
Capacitors



$$C = \frac{\epsilon_o A}{d}$$



$$C = \frac{2\pi\epsilon_o l}{\ln \frac{R_a}{R_b}}$$

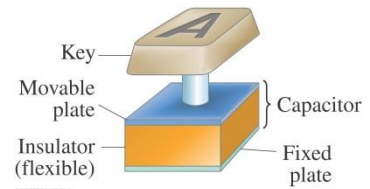


Giancoli pg 631 an application:

Many computer keyboard buttons are constructed of capacitors. What happens when the key is pushed down?

$$C = \frac{\epsilon_o A}{d}$$

Upper plates moves down and d is reduced, Increasing capacitance. Change in capacitance results in an electrical signal detected by circuit.



Some other applications:

1. Heart defibrillator (Giancoli pg 638)
2. Camera flash (Giancoli pg 636)
3. Window swiper
4. Capacitive touch screen.



<https://www.youtube.com/watch?v=qBbxSEp3-6o&t=4s>



<https://www.youtube.com/watch?v=COKBImkkJKw&t=2s>



<https://www.youtube.com/watch?v=ZrMltPK6iAw&t=2s>

Dielectric

[Knight pg 923, Ex 30.6]

The spacing between the plates of a $1.0\ \mu\text{F}$ capacitor is $0.0500\ \text{mm}$.

- What is the surface area of the plates?
- How much charge is stored on the plates if this capacitor is attached to a $1.5\ \text{V}$ battery?

See Giancoli pg 631 – high capacitance capacitors made from “activated” carbon.

Dielectric

We note that even for this typical capacitance of $1.0\ \mu\text{F}$, we need a large area. Earlier, we mentioned that there are two factors that determines the capacitance of capacitors – we have dealt with the geometry associated with the two plates, now we will see if altering the medium between the two plates helps. When we change the material, the permittivity changes,

$$\epsilon_o \rightarrow \epsilon = K\epsilon_o$$

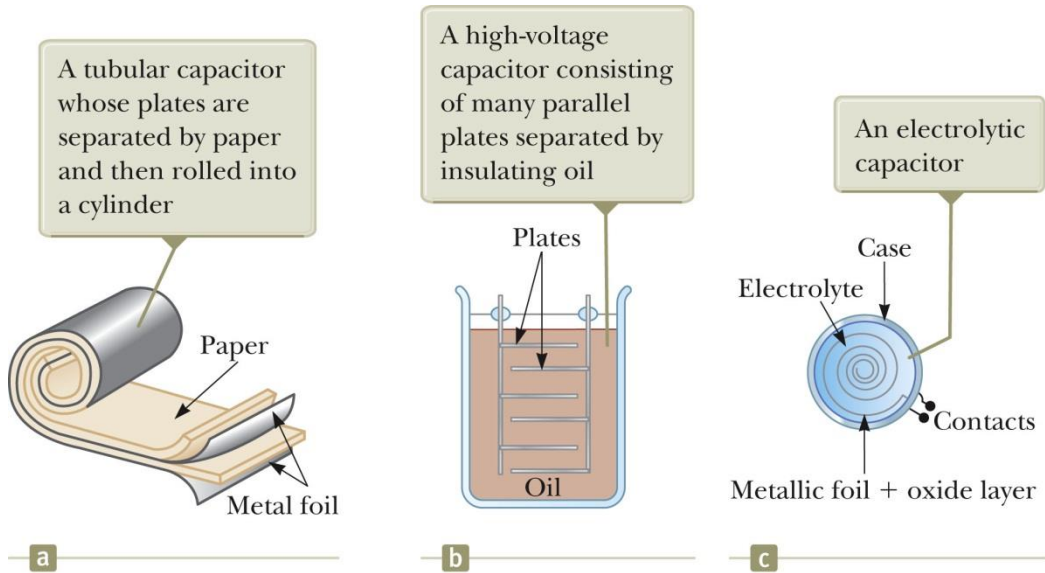
where K is the known as the dielectric constant which is determined by the material. Suppose now that Strontium titanate is used as the dielectric, determine the surface area required.

For SrTiO_3 , the maximum electric field that can exist before electrical breakdown is $8 \times 10^6\ \text{V/m}$. This quantity is known as the dielectric strength.

TABLE 24–1
Dielectric Constants (at 20°C)

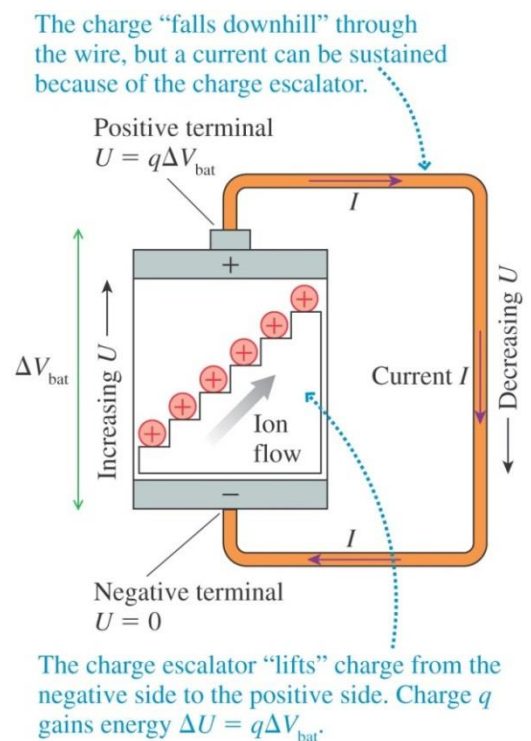
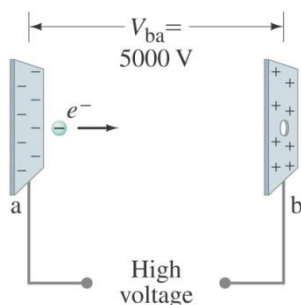
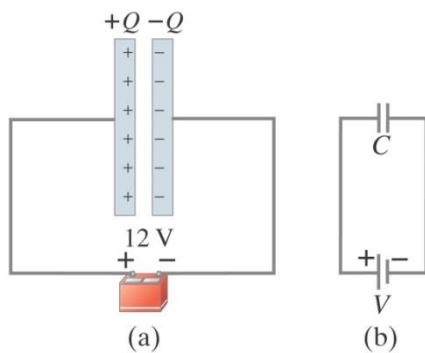
Material	Dielectric constant K	Dielectric strength (V/m)
Vacuum	1.0000	
Air (1 atm)	1.0006	3×10^6
Paraffin	2.2	10×10^6
Polystyrene	2.6	24×10^6
Vinyl (plastic)	2–4	50×10^6
Paper	3.7	15×10^6
Quartz	4.3	8×10^6
Oil	4	12×10^6
Glass, Pyrex	5	14×10^6
Porcelain	6–8	5×10^6
Mica	7	150×10^6
Water (liquid)	80	
Strontium titanate	300	8×10^6

Three commercial capacitor designs [Serway pg 755; Fig 26.14]



A dielectric increases the capacitance, maximum operating voltage and provide mechanical support between plates, allowing them to be close without touching.

Recap (Potential Energy, Potential and Charge)



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Energy stored in charged capacitor

The electrical energy stored in a capacitor can be used for various purposes – e.g. camera flash, defibrillators. How do we calculate this energy?

Earlier we learnt that the change in potential energy δU of a charge q moving through a potential difference δV has energy

$$\delta U = q\delta V$$

At first, transferring a charge q does not involve any work because the potential between the uncharged plates is zero. Then, as charges are moved across from one plate to the other, the potential difference between the plates increases and more work is required to bring the same quantity of charge across subsequently. This is depicted in the graph. The area under the graph gives the stored energy and is simply given by

$$U = \frac{1}{2} Q\Delta V.$$

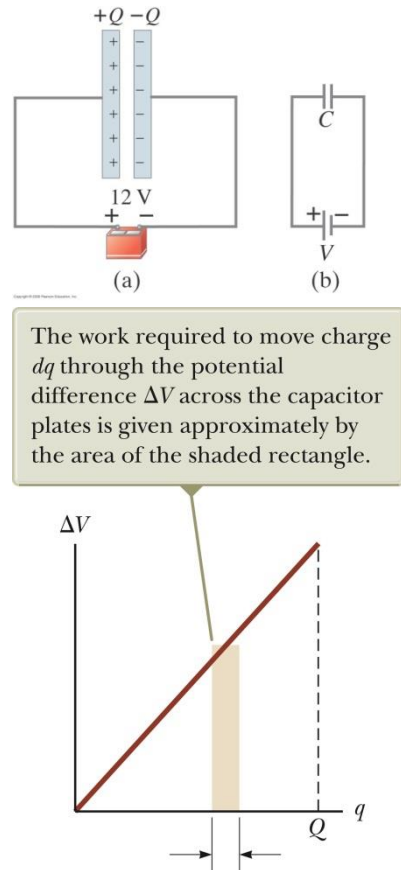
The same results can be obtained by integrating w.r.t dq

$$U = \int_0^Q \Delta V dq = \int_0^Q \frac{q}{C} dq = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2} C(\Delta V)^2 = \frac{1}{2} Q\Delta V$$

For parallel capacitors where $C = \frac{\epsilon A}{d}$ and $\Delta V = Ed$, we can write

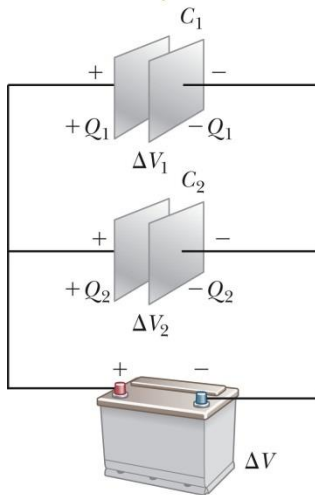
$$u_E = \frac{U}{Ad} = \frac{1}{2} \epsilon E^2.$$

The quantity u_E is known as the energy density – energy per unit volume. While this expression is derived using the parallel plate capacitor, it is also applicable for other sources of electric fields.



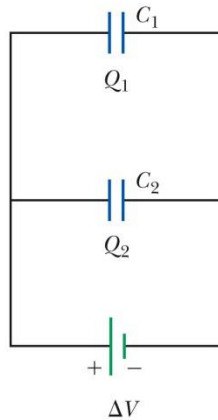
Combination of Capacitors (Parallel)

A pictorial representation of two capacitors connected in parallel to a battery



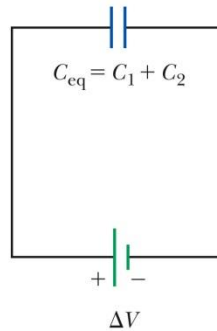
a

A circuit diagram showing the two capacitors connected in parallel to a battery



b

A circuit diagram showing the equivalent capacitance of the capacitors in parallel



c

Capacitor symbol

Battery symbol

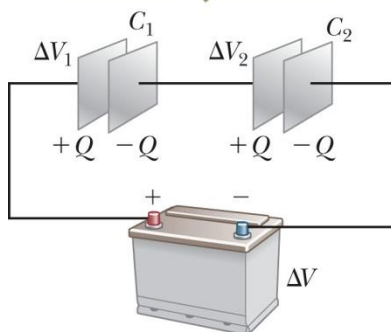
Switch symbol

Open

Closed

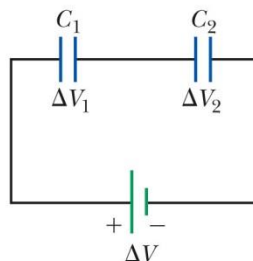
Combination of Capacitors (Series)

A pictorial representation of two capacitors connected in series to a battery



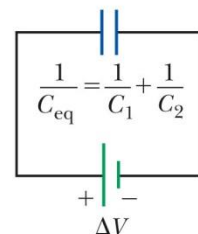
a

A circuit diagram showing the two capacitors connected in series to a battery



b

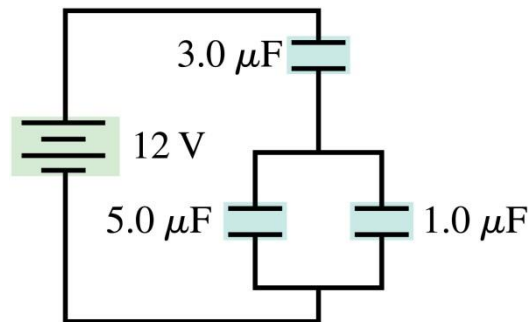
A circuit diagram showing the equivalent capacitance of the capacitors in series



c

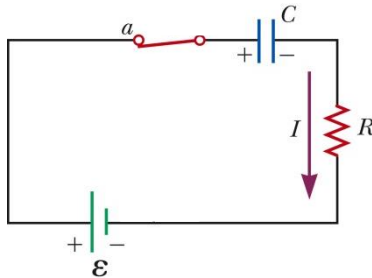
Example: [Knight pg 776 Ex 13.10]

- a. Find the equivalent capacitance of the combination of capacitors in the circuit.
- b. What charge flows through the battery as the capacitors are being charged?



R.C. circuits (Charging) [Derivation Non-examinable]

We have analyzed circuits involving resistors. Now we will also include capacitors in our analysis. One important difference is that the current will not be constant but we can still apply Kirchhoff's laws. We first apply Kirchhoff's loop law to the circuit:



Applying Kirchhoff's loop law, (see Giancoli pg 688)

$$\mathcal{E} = V_C + V_R$$

$$\mathcal{E} = \frac{Q}{C} + IR$$

The rate at which charge Q flows through the capacitor is $I = \frac{dQ}{dt}$ is to rate at which charge accumulates at the capacitor. Thus, we can write

$$\mathcal{E} = \frac{Q}{C} + R \frac{dQ}{dt}$$

Re-arranging, we get

$$\frac{dQ}{C\mathcal{E} - Q} = \frac{dt}{RC}$$

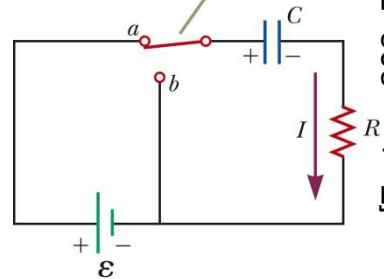
Integrating from $t = 0$ when $Q = 0$ to time $t = t'$ when charge Q' is on the capacitor

$$\begin{aligned} \int_0^{Q'} \frac{dQ}{C\mathcal{E} - Q} &= \frac{1}{RC} \int_0^{t'} dt \\ -\ln(C\mathcal{E} - Q) \Big|_0^{Q'} &= \frac{t'}{RC} - 0 \\ -\ln(C\mathcal{E} - Q') - (-\ln C\mathcal{E}) &= \frac{t'}{RC} \end{aligned}$$

Dropping the ' on Q' and t'

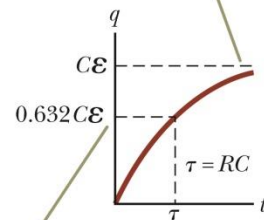
$$\begin{aligned} Q &= C\mathcal{E}(1 - e^{-\frac{t}{RC}}) \\ \Rightarrow V_C = \frac{Q}{C} &= \mathcal{E}(1 - e^{-\frac{t}{RC}}) \\ \Rightarrow I = \frac{dQ}{dt} &= \frac{\mathcal{E}}{R} e^{-t/RC} \end{aligned}$$

When the switch is thrown to position a , the capacitor begins to charge up.



[Text pg 808; Ex 28.16]

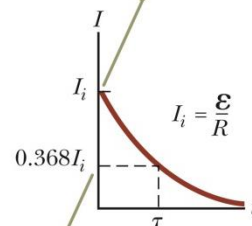
The charge approaches its maximum value $C\mathcal{E}$ as t approaches infinity.



After a time interval equal to one time constant τ has passed, the charge is 63.2% of the maximum value $C\mathcal{E}$.

a

The current has its maximum value $I_i = \mathcal{E}/R$ at $t = 0$ and decays to zero exponentially as t approaches infinity.



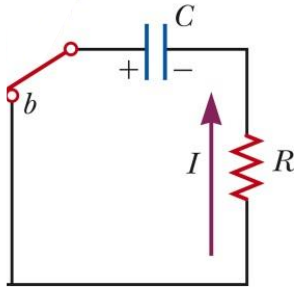
After a time interval equal to one time constant τ has passed, the current is 36.8% of its initial value.

b

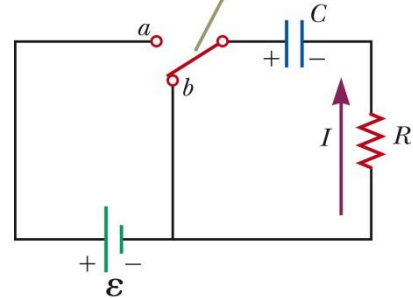
[Text pg 809; Ex 28.17]

R.C. circuits (Discharging) [Derivation Non-examinable]

Applying Kirchhoff's loop law to the circuit:



When the switch is thrown to position *b*, the capacitor discharges.



Applying Kirchhoff's loop law,

$$0 = V_c + V_R$$

$$0 = \frac{Q}{C} + IR$$

The rate at which charge Q flows through the resistor is $I = \frac{dQ}{dt}$. Thus, we can write

$$0 = \frac{Q}{C} + R \frac{dQ}{dt}$$

Re-arranging, we get

$$\frac{dQ}{Q} = -\frac{dt}{RC}$$

Integrating from $t = 0$ when $Q = Q_o$ to time $t = t'$ when charge Q' is on the capacitor

$$\int_{Q_o}^{Q'} \frac{dQ}{Q} = -\frac{1}{RC} \int_0^{t'} dt$$

$$\ln(Q) \Big|_{Q_o}^{Q'} = -\frac{t'}{RC} - 0$$

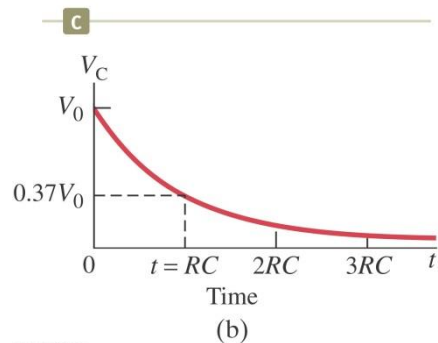
$$\ln(Q') - (\ln Q_o) = -\frac{t'}{RC}$$

Dropping the ' on Q' and t'

$$Q = Q_o e^{-t/RC}$$

$$\Rightarrow V_c = \frac{Q}{C} = V_o e^{-\frac{t}{RC}}$$

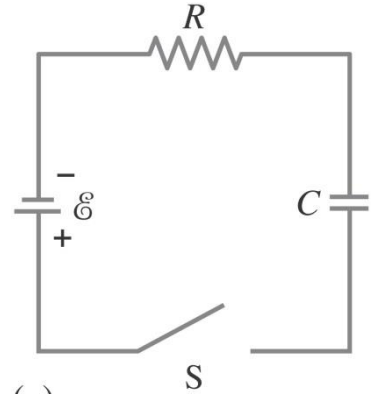
$$\Rightarrow I = \frac{dQ}{dt} = -\frac{Q_o}{RC} e^{-\frac{t}{RC}} = -I_o e^{-\frac{t}{RC}}$$



<https://www.youtube.com/watch?v=OEL5laB3hfU&t=4s>

Example: [Giancoli pg 689 Ex 26-11]

The capacitance in the circuit shown is $C = 0.30 \mu\text{F}$, the total resistance is $20 \text{ k}\Omega$, and the battery emf is 12 V . Determine (a) the time constant, (b) the maximum charge the capacitor could acquire, (c) the time it takes for the charge to reach 99% of this value, (d) the current I when the charge Q is half its maximum value, (e) the maximum current, and (f) the charge Q when the current I is 0.20 its maximum value.



d. $\mathcal{E} = V_R + V_C$

f. $\mathcal{E} = V_R + V_C$